



Kvaerner Oil & Gas

Project : Huldra EPC Wellhead Platform Topsides

Title : Stystem Control Diagram, Air Compressor B Sequences

Doc. No : C025-W-H063-J-*E-006-01

KOG Project No : C131

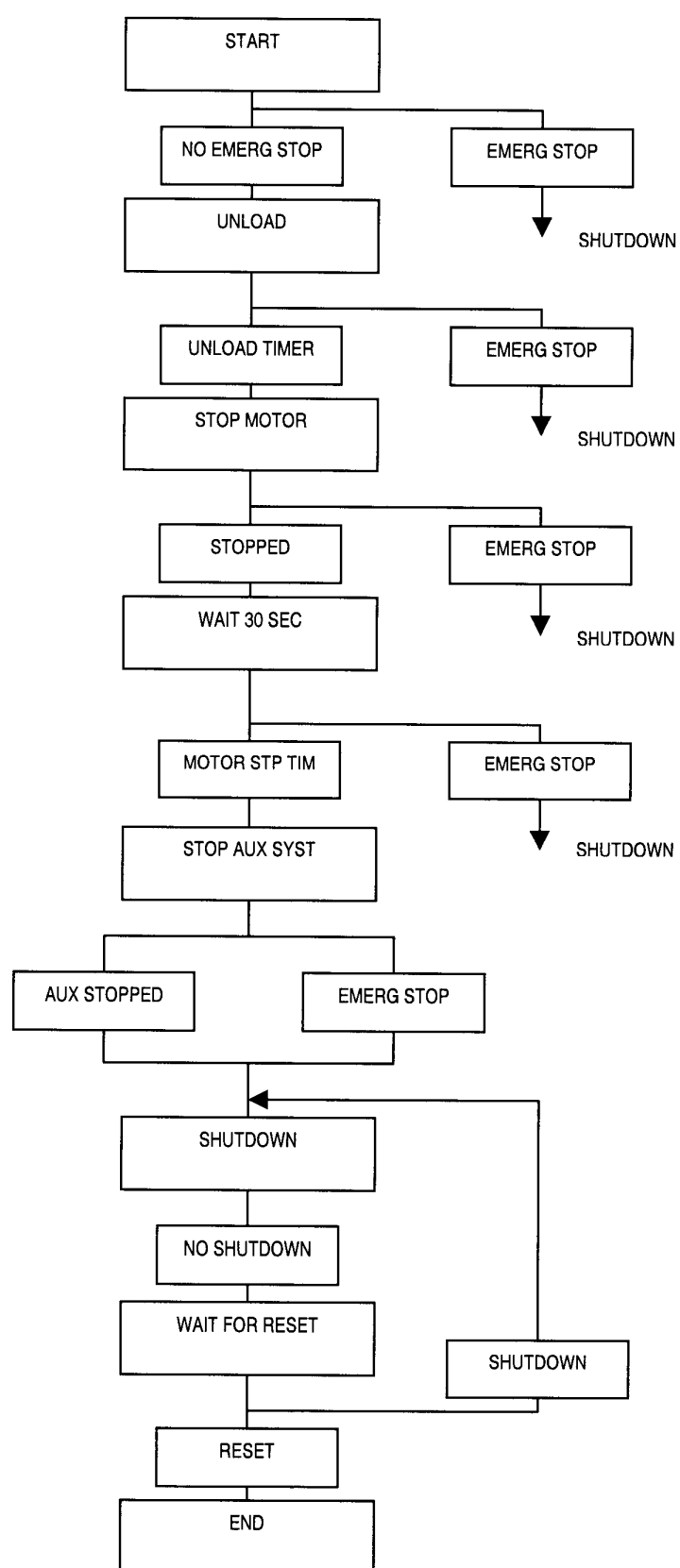
Statoil Project No : C025

A2	12.09.03	As Built AO600126	JB	SMB/RT	DB
A1	30.09.01	AS BUILT	EiL	<i>CE/S</i>	<i>PD</i>
0	20.03.01	ISSUED FOR CONSTRUCTION	Buk		
01D	30.11.00	ISSUED FOR DESIGN	Buk		
Revision	Date	Revision Description	Prepared	Checked	Approved

INPUTS		
Document Number	Title	Status
C025-W-H063-J-SX-001 Comp Air: E7100-C-02	Functional description air distribution system 63 Air Cooled DRYCLON CONTROLLER, Chemitrol/K.O.G./STATOIL (Huldra)	04D A2

REVISIONS & HOLDS	
Revision	Description of Changes & Holds
01D	Issued for design
0	Issued for construction – Statoil comments incorporated
A1	As Built
A2	As Built AO600126





COMPRESSOR B SEQUENCE

Phase	Step	Actions	Outputs to	Next step	Wait for (before entering next state)	Jump condition.	Jump to step	Remarks
START	START	N.A.	N.A	NEXT	N.A.	N.A.	N.A.	**
Start inhibit	1	All equipment de-energized	Stop signal to: 63-KC 50B 63-KE 51A 63-KE 51B 63-PE 51 Close signal to: 63-XV 2052 63-HV 2067 Set 63-KE 51B to manual	NEXT	Available and BCL from equipment: 1: 63-KC 50B .XE AND 2: 63-KE 51A.XE AND 3: 63-KE 51B.XE AND 4: 63-PE 51.XE AND 5: 63-KC 50B.BCL AND 6: 63-KE 51A.BCL AND 7: 63-PE 51.BCL AND = TRUE	N.A.	N.A.	*
Ready for start	2	All equipment de-energized	Stop signal to: 63-KC 50B 63-KE 51A 63-KE 51B 63-PE 51 Close signal to: 63-XV 2052 63-HV 2067 Set 63-KE 51B to manual	NEXT OR JUMP 1 OR JUMP 2	Start command from OS or logic: 63-HS 9001.Y0 OR (63-PI 2000. BXL AND Z253.Y) OR (63-HS 9000L.Y AND Z254.Y) =TRUE	JUMP 1: Equipment not available: 1: 63-KC 50B .XE AND 2: 63-KE 51A.XE AND 3: 63-KE 51B.XE AND 4: 63-PE 51.XE AND 5: 63-KC 50B.BCL AND 6: 63-KE 51A.BCL AND 7: 63-PE 51.BCL AND = FALSE JUMP 2: Stop command from OS 63-HS 9001.Y1 = TRUE	JUMP 1: (Start inhibit) JUMP 2: (Go to SD sequence)	*
Standby	3	Aux. equipment de-energized	Stop signal to: 63-KE 51A 63-KE 51B 63-PE 51 Close signal to: 63-XV 2052 63-HV 2067 Set 63-KE 51B to manual	NEXT OR JUMP	Low pressure: 63-PI 2000.BXL=TRUE	JUMP 2: Stop command from OS: 63-HS 9001.Y1=TRUE	2 (Ready for start)	*
Start aux.syst	4	Start oil pump, open discharge valve, start fan 1 and set fan 2 to auto.	Start signal to: 63-KE 51A AND 63-KE51B 63-PE 51 Open s,ignal to: 63-HV 2067 Set 63-KE 51B to auto	NEXT OR JUMP	Aux. Syst confirmed started: 63-PE 51.BCH AND 63-KE 51A OR 63-KE51B AND 63-PI 2055.BXH = TRUE	Stop command from OS: 63-HS 9001.Y1=TRUE	2 (Ready for start)	*
Start motor	5	Inhibit vibration alarms Start Compressor B main drive	63-YAH 2068A/B.FU and 63-YAHH 2068A/B.FU = TRUE (via Z318) Start sinal to: 63-KC 50B	NEXT OR JUMP	Motor running: 63-KC 50B.BCH = TRUE	Stop command from OS: 63-HS 9001.Y1=TRUE	12 (Stopping not available)	*
Load delay	6	Start Load Delay Timer– 2 sec	Start timer T1163 (Z163.X)	NEXT OR JUMP	Load delay expired: T1163(Z163.Y) = TRUE	Stop command from OS: 63-HS 9001.Y1=TRUE	12 (Stopping not available)	
Offload	7	De-energize load valve Turn off inhibit of vibration alarm Start Run On Timer – 15 m	Close signal to: 63-XV 2052 63-YAH 2068A/B.FU and 63-YAHH 2068A/B.FU = FALSE (via Z318) Start timer T1164 (Z164.X)	NEXT OR JUMP 1 OR JUMP 2	Low pressure and discharge valve confirmed open and, load timer or reload timer: 63-PI 2000. BXL AND 63-HV 2067.BCH AND (T1163(Z163.Y) OR T1166(Z166.Y)) = TRUE	JUMP 1: Run On Timer: T1164(Z164.Y) = TRUE JUMP 2: Stop command from OS: 63-HS 9001.Y1=TRUE	JUMP 1: 9 (Stopping available) JUMP 2: 11 (Unloading)	*

Onload	8	Energize load valve Start reload timer on exit from step.	Open signal to: 63-XV 2052 Start timer T1166 (Z166.X) on exit from step	JUMP 1 OR JUMP 2	N.A.		JUMP 1: High press or Discharge Valve NOT confirmed open: 63-PI 2064.BXH = TRUE OR 63-HV 2067.BCH = FALSE	JUMP 1: 7 (Offload) JUMP 2: 11 (Unloading)	*
Stopping available	9	Inhibit vibration alarms Stop Compressop B main drive	63-YAH 2068A/B.FU and 63-YAHH 2068A/B.FU = TRUE (via Z318) Stop signal to: 63-KC 50B	NEXT	Motor stopped: 63-KC 50B.BCL = TRUE		JUMP 2: Stop command from OS: 63-HS 9001.Y1=TRUE N.A.		*
Motor Stop Timer a	10	Start Motor Stop Timer – 30 sec Turn off inhibit of vibration alarm on exit from step	Start timer T1165 (Z165.X via Z330) 63-YAH 2068A/B.FU and 63-YAHH 2068A/B.FU = FALSE (via Z318)	JUMP 1 OR JUMP 2	N.A.	JUMP 1: Motor stop timer expired: T1165(Z165.Y) = TRUE AND 63-HS 9001.Y1 = FALSE	JUMP 1: Motor stop timer expired: T1165(Z165.Y) = TRUE AND 63-HS 9001.Y1 = FALSE	JUMP 1: 3 (Standby) JUMP 2: 12 (Stopping not available)	
Unload	11	De-energize load valve and start Start Unload Timer – 8 sec.	Close signal to: 63-XV 2052 Start timer T1167 (Z167.X)	NEXT OR JUMP	Unloading time expired: T1167(Z167).Y = TRUE	JUMP 2: Stop command from OS: 63-HS 9001.Y1=TRUE	Start command from OS: 63-HS 9001.Y0=TRUE	7 (Offload)	*
Stopping not available	12	Inhibit vibration alarms Stop Compressop B main drive	63-YAH 2068A/B.FU and 63-YAHH 2068A/B.FU = TRUE (via Z318) Stop signal to: 63-KC 50B	NEXT	Motor stopped: 63-KC 50B.BCL = TRUE	N.A.			*
Motor Stop Timer b	13	Start Motor Stop Timer – 30 sec Turn off inhibit of vibration alarm on exit from step	Start timer T1165 (Z165.X via Z330) 63-YAH 2068A/B.FU and 63-YAHH 2068A/B.FU = FALSE (via Z318)	JUMP 1 OR JUMP 2		JUMP 1: Start command from OS: 63-HS 9001.Y0=TRUE JUMP 2: Motor stop time expired: T1165(Z165.Y) = TRUE	JUMP 1: Start command from OS: 63-HS 9001.Y0=TRUE JUMP 2: Motor stop time expired: T1165(Z165.Y) = TRUE	JUMP 1: 9 (Stopping available) JUMP 2: 1 (Start inhibit)	
Go to SD sequence	14	Start 63-SD_COMP_B	Start command to 63-HS 9001L	END	N.A	N.A	N.A	N.A	
End	15								

* When a start/ stop or open/close command is given to an object, an auto command is also given.
** The sequence is started automatically upon start of the PLC. It will run as long as the shutdown sequence is not running. I.e. the compressor sequence is aborted when he shutdown sequence is started. When the shutdown sequence is terminated the compressor sequence is restarted.

Chart Properties\ Operating Parameters:

Defaults:

Step Control:	T
Operating Mode:	Auto
Command output:	Yes
Cyclic operation:	Yes
Time monitoring:	No
Start Chart:	
Autostart:	Yes
Use default settings	
when SFC start:	Yes

SHUTDOWN SEQUENCE COMPRESSOR B

Phase	Step	Actions	Outputs to	Next step	Wait for (before entering next step)	Jump condition	Jump to step	Remarks
START	START	N.A.	N.A.	NEXT OR JUMP	NOT Emergency stop: Z232.OUT = FALSE	Emergency Stop Activated: Z232.OUT = TRUE	5 (Shutdown)	The sequence is started by Z171.OUT See SCD C025-W-H063-J-*E-003-01
Unload	1	Stop Comp_B sequence De-energize load valve Start Unload Timer – 2 sec	Stop command to 63-HS 9001H Close signal to 63-XV 2052 Start timer T1168 (Z168.X)	NEXT OR JUMP	Unload timer - shutdown : Z168.Y	Emergency Stop Activated: Z232.OUT = TRUE	5 (Shutdown)	*
Stop motor	2	Inhibit vibration alarms Stop Compressop B main drive	63-YAH 2068A/B.FU and 63-YAHH 2068A/B.FU = TRUE (via Z318) Stop signal to 63-KC 50B	NEXT OR JUMP	Main drive motor confirmed stopped: 63-KC 50B.BCL = TRUE	Emergency Stop Activated: Z232.OUT = TRUE	5 (Shutdown)	*
Motor Stop Timer- Wait 30 sec.	3	Start Motor Stop Timer – 30 sec Turn off inhibit of vibration alarm on exit from step	Start timer T1165 (Z165.X) (via Z330) 63-YAH 2068A/B.FU and 63-YAHH 2068A/B.FU = FALSE (via Z318)	NEXT OR JUMP	Motor Stop Timer : Z165.Y	Emergency Stop Activated: Z232.OUT = TRUE	5 (Shutdown)	
Stop Aux. syst	4	Stop oil pump, close discharge valve, stop fans and set fan 2 to manual	Stop signal to: 63-PE 51 63-KE 51A 63-KE 51B Close signal to: 63-HV 2067 Set 63-KE 51B to manual	NEXT	Aux. system confirmed stopped or emergency stop: 63-KE 51A.BCL and 63-KE 51B.BCL and 63-PE 51.BCL and 63-HV 2067.BCL = TRUE OR Z232.OUT = TRUE	N.A.	N.A.	*
Shutdown	5	Stop Comp_B sequence	Stop command to 63-HS 9001H	NEXT	No emergency stop and no Shutdown command: Z171.OUT AND Z232.OUT = FALSE	N.A.	N.A.	*
Wait for reset	6	Reset command from OS	N.A.	NEXT OR JUMP	63-HS 9001.Y2 = TRUE	Emergency stop or Controlled shutdown command: Z232.OUT OR Z171.OUT = TRUE	5 (Shutdown)	
End	END	Start Comp_B sequence	Start command to 63-HS 9001H	END	N.A.	N.A.	N.A.	*

* When a start/ stop or open/close command is given to an object, an auto command is also given.

Chart Properties\ Operating Parameters:

Defaults:

Step Control:

Operating Mode:

Command output:

Cyclic operation:

Time monitoring:

Start Chart:

Autostart:

Use default settings

when SFC start:

T

Auto

Yes

No

No

Yes

Yes

Yes